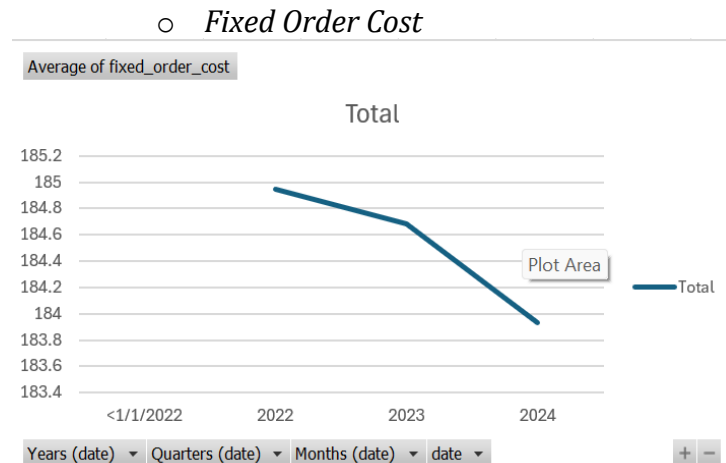
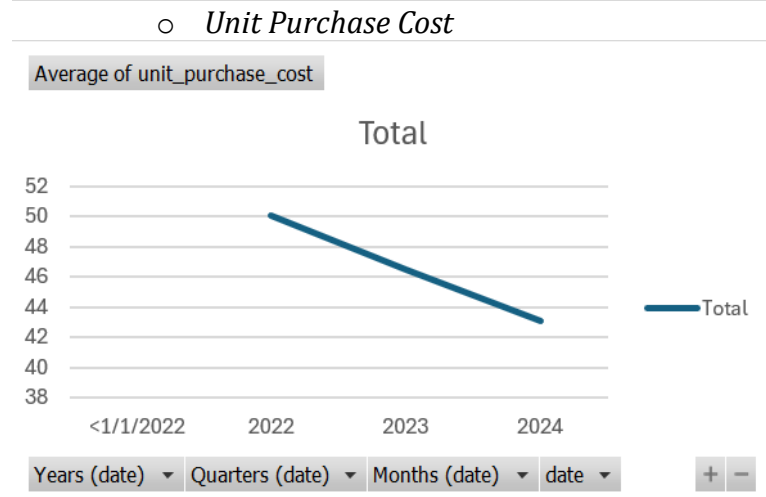
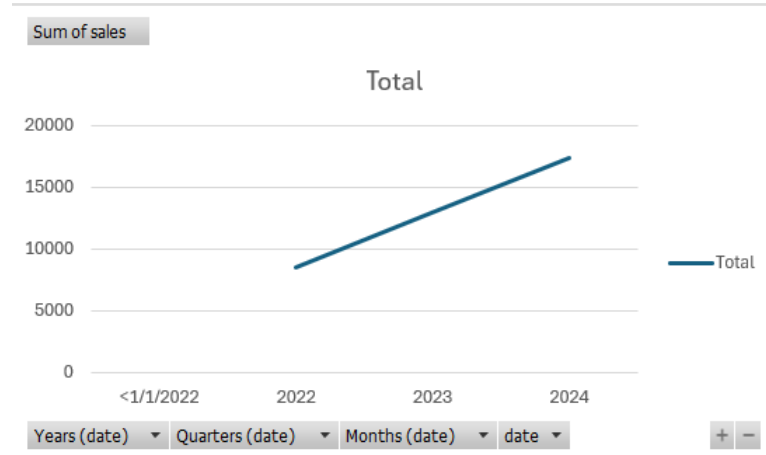


Module 11 – EOQ

Exploratory Data Analysis

- Make line graphs showing the following data over time:
 - Sales



- Use a forecast method to determine annual demand for 2025 to use for our model
 - o Annual Demand using Naïve is 17365

Model Formulation

Decision Variables

D = annual demand

C = unit purchase cost

S = fixed cost of placing order

I = cost of holding inventory

Q = order quantity

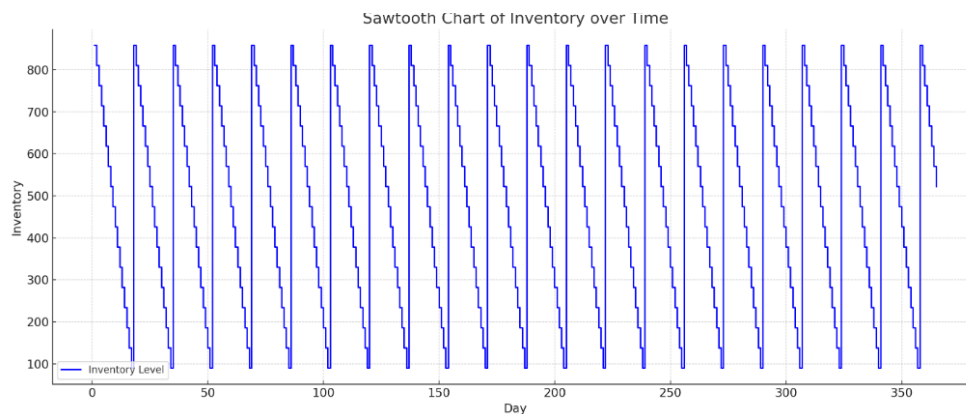
B = Backorder

A = Shortage cost

$$\text{Objective Function} = TC(Q, b) = D/Q(S) + H/2 * (Q-b)^2/Q + A/2 * b^2/Q$$

$$\text{Constraints} = Q \geq 0 \text{ and } B \geq 0$$

Model Optimized for Minimizing Costs with Optimal Order Quantity

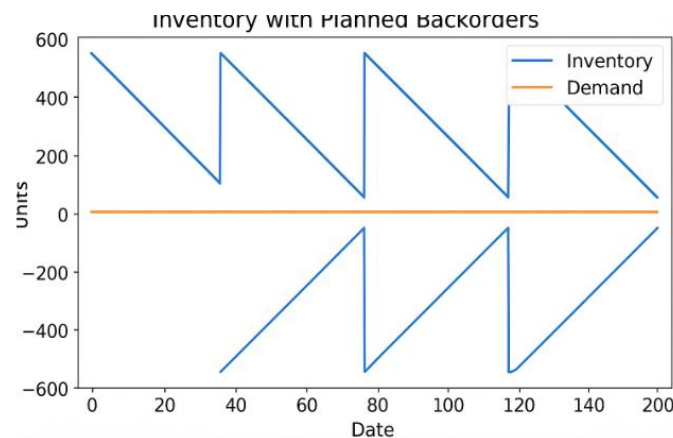


This sawtooth chart represents inventory levels over time assuming a constant demand of 48 units per day, a restock to 906 units when inventory drops below the daily demand. Each day 48 units are consumed which causes inventory to decrease, this continues until inventory can no longer satisfy another day of demand. Once inventory is restocked it reaches the max level of 906, which is the upward spike.

Model with Stipulation

D	Annual Demand	17365
C	Cost per Unit	\$ 43
S	Cost per Order	\$ 183
A	Shortage Cost	\$ 20
B	Backorders	478
Q	Starting EOQ	571
I	Holding Cost	18%
	Order Quantity	571
	Planned BO	\$ 4,004
	Cost of Ordering	\$ 5,562
	Inventory Cost	\$ 779
	Total Cost	\$ 757,040
	Total relevant Cost	\$ 4,004

$$\text{Total Relevant Cost} = \frac{D}{Q}S + \frac{(Q - b)^2}{2Q}C_i + \frac{b^2}{2Q}A$$



Including planned backorders in inventory management can be a strategic move to optimize costs and balance service levels. Instead of always keeping enough stock on hand, businesses may allow temporary stockouts, accepting that some customers are willing to wait for their orders. This approach reduces holding costs, which are often significant, especially when inventory ties up capital or space. It's particularly useful when demand is predictable, or when carrying inventory is more expensive than the cost of occasionally delaying fulfillment. The updated sawtooth chart visually represents this strategy: inventory starts at a positive level (571 units), declines over time due to demand, and then dips below zero (to -478 units), indicating backorders. When a new order arrives, the inventory jumps back up, covering those

unfulfilled orders and resetting the cycle. This planned fluctuation allows businesses to operate more efficiently while maintaining acceptable customer service levels.